Virtual Lab Pre-Lab: Empirical Formula of Magnesium Oxide

Name: _____ Date: _____ Period: _____

Goal: Use experimental data to determine the ratio at which two or more elements combine.

Introduction: In the real lab you will react magnesium metal with the oxygen gas in the air. The goal is to find the ratio at which magnesium atoms and oxygen atoms combine to form a compound. Before you do it for real, you will practice the setup and the calculations in a virtual lab. Run the simulation with this sheet beside you. **Write every reading, observation and calculation on this sheet.** The simulation will tell you which box to fill in.

Question 1. Write the chemical reaction you expect to happen between magnesium and oxygen gas. Balance it.

_____ Mg + _____ O_2 → _____

Data table

Box	What you measure	Your reading
1	Mass of crucible + lid	g
2	Mass of Mg	g
3	Mass of crucible + lid + Mg	g
4	Mass of crucible + lid + product, after the reaction	g
5	Mass of oxygen added (Box 4 − Box 3)	g

Observations (the flame, the magnesium, the white product, what you see and smell when you add water):

_____

_____

_____

Procedure (what you do in the simulation)

1) **Set up the bench.** Place the crucible in the clay triangle on the ring stand, with the Bunsen burner underneath.
2) **Mass the clean, dry crucible and lid.** Record it in **Box 1**.
3) **Mass the piece of magnesium.** Record it in **Box 2**. Put it in the crucible, then mass the crucible + lid + Mg and record it in **Box 3**. Check: Box 1 + Box 2 should equal Box 3.
4) **Heat the magnesium with the Bunsen burner.** Turn the bottom of the burner until the tip of the flame touches the bottom of the crucible. The lid goes on for you, with a slight crack, once the Mg glows. Write what you see in Observations.
5) **Once the Mg stops glowing, add 6 to 7 drops of DI water** and reheat to boil the water off. Write what you see and smell.
6) **Once the product is dry, turn the heat off and cool for 5 minutes.** The simulation speeds up the clock. Do the **While Cooling** questions.
7) **Mass the crucible + lid + product.** Record it in **Box 4**. Then calculate **Box 5**.

While Cooling

1. Calculate the number of moles of Mg used in this experiment. Show your work with units.

2. Calculate the number of Mg atoms that were in the ribbon.

3. Do you expect the mass of the crucible + lid + product to be smaller, larger, or the same as before the reaction? Explain your answer.

After Cooling

1. Calculate the moles of oxygen present in the product. (Use 16 g/mol for the molar mass of oxygen, not 32 g/mol.)

2. Compare the number of moles of magnesium to the moles of oxygen. What do you notice?

3. Based on your data, what formula did the experiment lead to?

Look closer: magnesium and nitrogen (use the magnifier)

Why do we add water?

Air is about 78% nitrogen gas, N_2. When magnesium burns it reacts with some of the nitrogen as well as the oxygen. It makes a pale yellow-green powder called magnesium **nitride**. If any nitride were left, the product would not be only MgO, and the mass you measure would not match what you calculate for MgO.

That is why you add water after the Mg stops glowing. The water reacts with magnesium nitride and **converts it to magnesium hydroxide**, $Mg(OH)_2$, and ammonia gas. When you reheat, the water boils off and the magnesium hydroxide changes into MgO, so everything left in the crucible is MgO.

N1. Magnesium forms Mg^{2+} and nitrogen forms N^{3-}. Write the formula for magnesium nitride and explain how the charges decide the subscripts.

N2. Balance the reaction of magnesium with nitrogen gas: _____ Mg + _____ N_2 → _____ Mg_3N_2

N3. Balance the reaction of the water with magnesium nitride: _____ Mg_3N_2 + _____ H_2O → _____ $Mg(OH)_2$ + _____ NH_3

N4. One product in N3 is a gas with a sharp smell. Which one is it? What happens to it in the crucible?

N5. Balance the change that happens when you reheat: _____ $Mg(OH)_2$ → _____ MgO + _____ H_2O. In one sentence, why does the water step let us treat the final product as all MgO?

What if…? (use the "what if" buttons in the simulation)

For each mistake, circle what would happen to your numbers and explain in one sentence. Assume everything else is done correctly.

What goes wrong	Mass of crucible + lid + product would be:	The calculated O : Mg ratio would be:	Why?
a) The lid is on **too wide**: white smoke escapes	too high / too low	too high / too low	
b) The lid is on **too tight**: no gap	too high / too low	too high / too low	
c) You **skip the water** step	too high / too low	too high / too low	